High-Reach Mechanical Scraper: Engineering an Accessible Solution for Spotted Lanternfly Egg Mass Removal

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SLF Egg Masses Reside Out of Reach, Requiring Ground-Based Mechanical Removal

- SLF egg masses are how the species continues to spread and can appear on any surface and are best removed by scraping and crushing
- ~90% of masses are deposited at least 8–10 ft above ground — out of reach without a ladder (nj.gov)
- Our high-reach, pole-mounted scraper enables safe, single-person removal from the ground

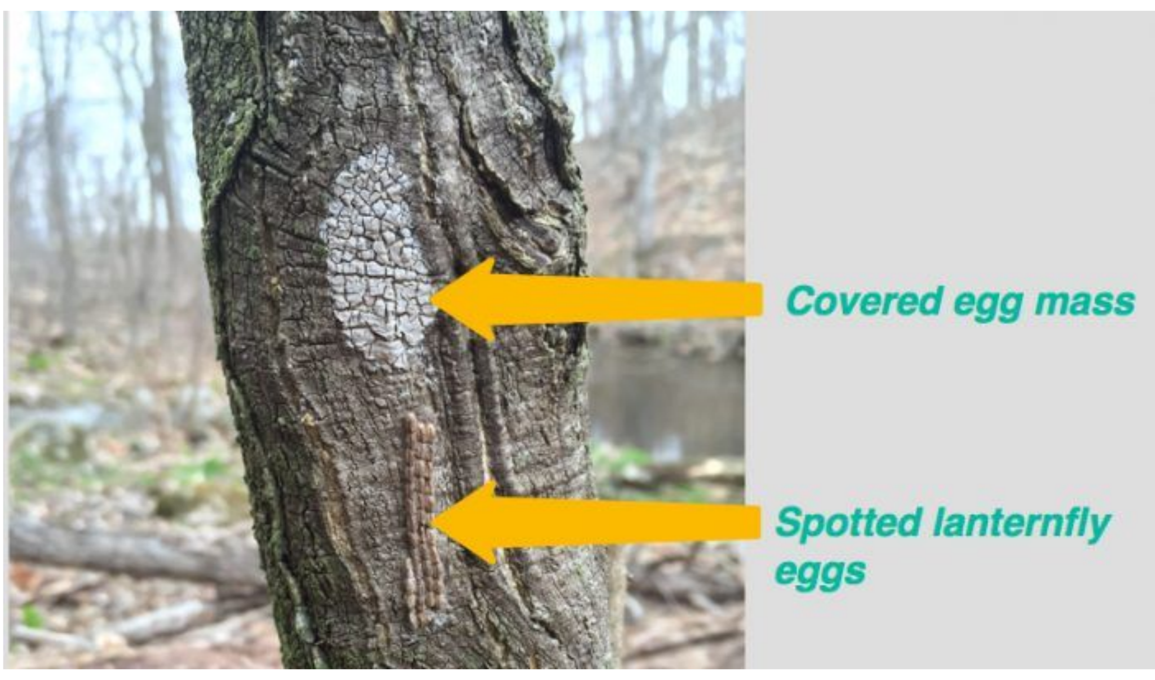

Figure 1. SLF Egg Mass Adhesion and Placement. SLF egg masses are highly camouflaged and adhere tightly to rough surfaces like bark. Over 90% of these masses are deposited 8-10 feet above ground .(The Spotted Lanternfly Project]

A Telescoping Mechanism Safely Extends Reach by 6 Feet

Design Goal: The lightweight, adjustable pole is able to extend a scraper and collection bin up to 6 feet above a user's height, allowing one to reach a much greater fraction of egg masses

Success Criteria:

- Scraping efficacy: >90% removal in 3 motions
- Total weight: Not exceeding 10 pounds
- Scraper Edge Durability: <5mm loss after 40 cycles
- Operational reach: scraping efficacy meets same standards at height of 8-10 feet

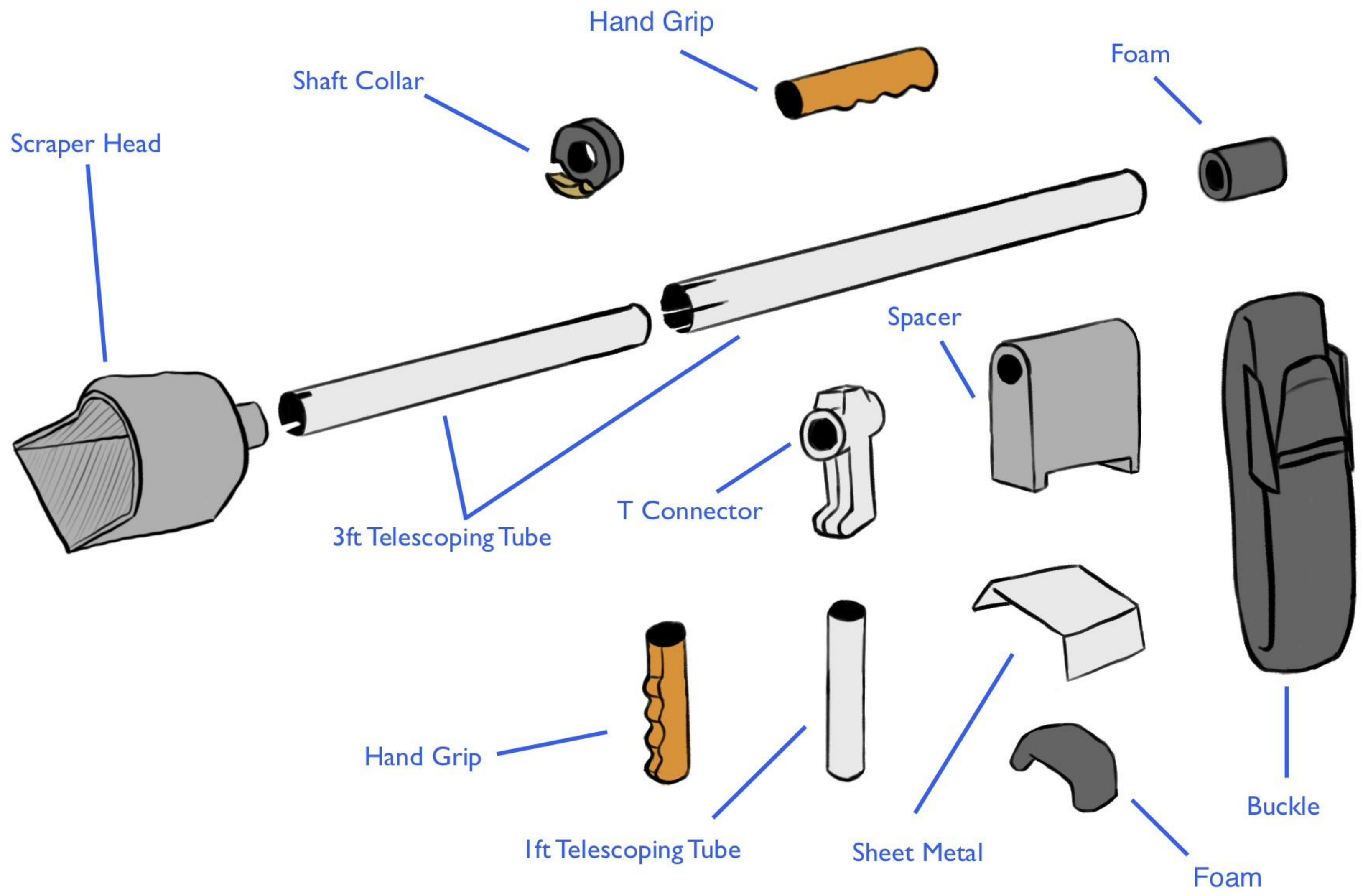

Figure 2. Exploded CAD Assembly of Final Prototype. The system features a telescoping pole, a custom head, and an ergonomic handle. To manage high torque generated by scraping, the base incorporates a forearm brace.

Rigorous Testing Validated Scraping Efficacy, Durability, and Ergonomic Reach

Scraping Efficacy
10 model egg masses (made of clay) were placed on smooth walls at varying heights, ranging from 29-200 cm from the floor.

Scraper Durability
To simulate rough bark, the scraper was drawn across sandpaper for two sets of 20 scraping cycles. Material loss per cycle was calculated.

Weight and Reach
Adjust length and weight to optimize user comfortability. Early prototype calculations found weight should not exceed 10 pounds for manageable moment.

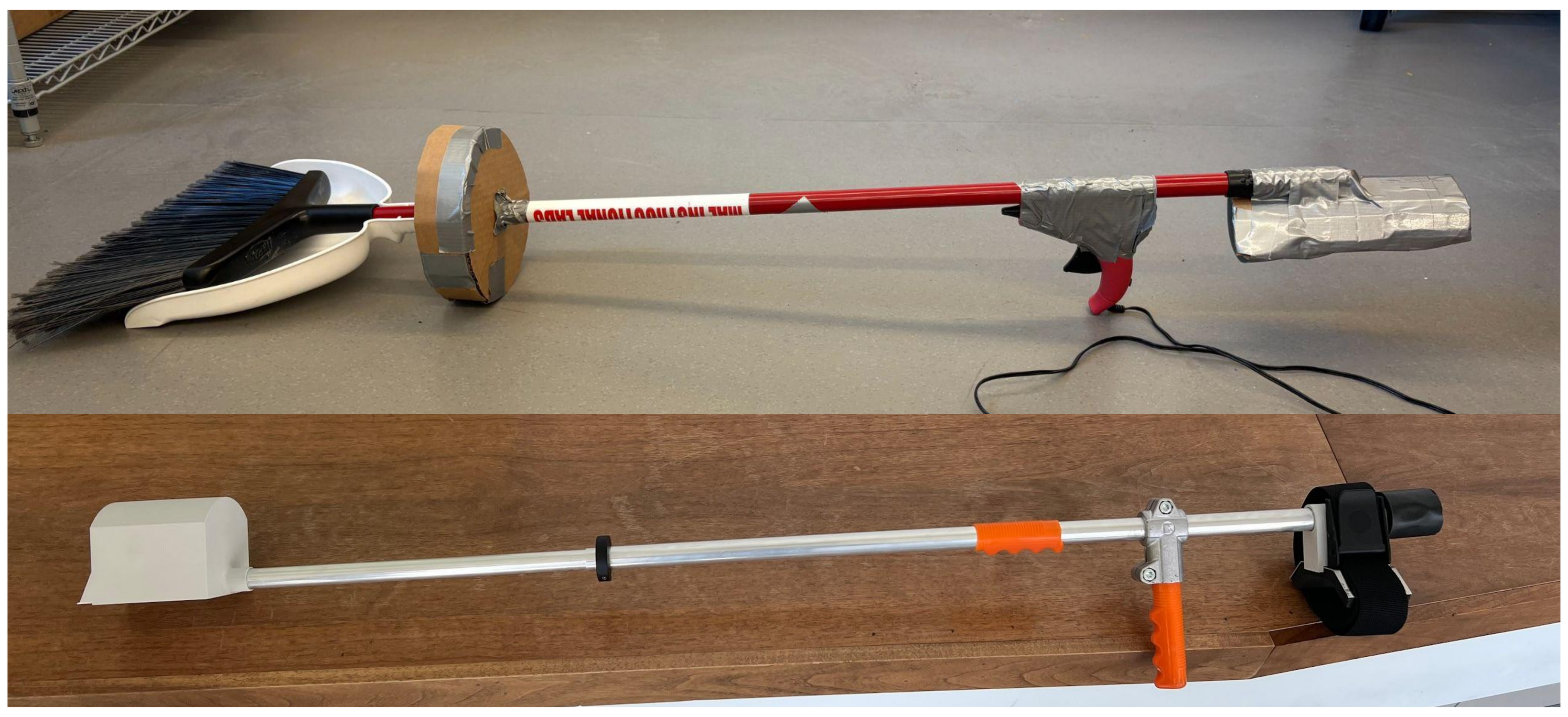

Figure 3. Initial and Final Prototype Side by Side Comparison. The first prototype was made with scrap materials while the final prototype incorporates the same design elements with professional grade materials.

The Final Prototype Exceeds All Quantitative Success Criteria

Durability
There was no measurable loss in the thickness of the metal edge, though it became slightly dulled.

Scraping Efficacy
The scraper removed >90% of clay egg mass replicas in minimal strokes. The clay model is heavier and stickier than real SLF egg masses, so the actual performance should be better.

Weight
The device weighs 2.85 lbs, making the moment arm at any height manageable.

Operational Reach
Max extension reaches well over 10 feet while maintaining efficacy standards.

Figure 4. First Functional vs Final Prototype Scraper Efficacy and Durability Comparison. The final prototype proved to be more effective on smooth and rough surfaces, while also maintaining its edge unlike the first prototype tested.

The Prototype is Field-Ready and Requires Only Minor Geometric Iterations

Findings & Conclusions

- **Criteria Validated:** Meeting all quantitative success criteria confirms the prototype successfully achieved the primary design goals.
- **Ergonomics & Stability:** The telescoping mechanism proved mechanically robust while maintaining user-friendly ergonomics under operational loads.
- **Operational Envelope:** The tool's maximum extension provides a reach capable of accessing nearly all egg mass locations within standard vineyards, with secondary applicability in forested environments.
- **Material Upgrades:** Integrating a sheet metal scraping edge significantly enhanced both detachment efficacy and long-term durability compared to the initial plastic iterations.

Future Considerations & Next Steps

- **Live Field Trials:** Execute testing on live SLF egg masses in outdoor agricultural conditions to acquire real-world environmental efficacy data.
- **Modular Tooling:** Develop interchangeable attachments (e.g., localized chemical sprayers or wire brushes) to expand the device's functional scope.
- **Geometric Iterations:** Design alternative scraper head profiles optimized for highly irregular host-plant surfaces and varied environments.

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